A Hopf Algebra on Proof Trees over a Nonmonotonic Consequence Relation

Abstract

We construct a pre-Lie algebra on proof trees for a multisuccedent sequent calculus equipped to a primitive base consequence relation which is not assumed to be monotonic. The sequent calculus naturally gives rise to decorated and labelled rooted trees, while the inadmissibility of standard Weakening rules gives rise to a constrained grafting operation on trees. We define a proof-theoretically meaningful equivalence relation which identifies proof-trees differing only in inessential or bureaucratic scheduling or reassociation choices. Passing to the induced quotient we show that there is a clean separation between cancelling and genuinely asymmetric contributions to the associator, yielding a pre-Lie identity for proof-tree composition. This extends to a Hopf algebra by the Oudom-Guin construction. This suggests that Hopf-algebraic methods on proof trees are a potential tool for structural proof theory, in the sense that the presence or absence of structural rules controls the existence and form of the algebraic structure.

1 Introduction

The aim of this paper is to describe a proof-theoretically meaningful algebraic setting in which monotonicity of consequence and the familiar structural rules are not built in from the outset. This allows structural rules to be analysed as constraints on algebraic structures.

Our proof calculus is the Non-Monotonic Multi-Succedent sequent calculus NMMS introduced by Kaplan (2018). Nonmonotonicity of consequence here is represented proof-theoretically by the failure of the standard structural rules of Weakening:

$$\frac{\Gamma \mid\!\sim \Delta}{\Gamma, A \mid\!\sim \Delta}\ \textit{(WL)} \qquad\qquad \frac{\Gamma \mid\!\sim \Delta}{\Gamma \mid\!\sim A, \Delta}\ \textit{(WR)}$$

These rules are *not* included in $NMMS(\mid\!\sim_0)$. Their absence is not a syntactic restriction but a proof-theoretic necessity: in a nonmonotonic setting, adding a formula to the context of a derivable sequent can defeat the underlying base inference.

Structural rules such as Weakening have of course been studied extensively in the context of substructural logics. Linear logic in particular controls extensions of contexts and multiplicity of hypotheses via constraints on the Weakening and Contraction. Linear logic also possesses the Cut Elimination property which is most desirable from the point of view of computational content. NMMS is a very different setting – here the sequent calculus is equipped on a primitive base consequence relation $\mid\!\sim_0$, which serves to encode non-logical patterns of defeasible *material inference* of the sort studied by Brandom and collaborators in the *inferentialist* analysis of meaning in natural language. Nonmonotonic consequence relations of this sort fail in general to admit the Cut rule:

The inadmissibility of Cut entails that NMMS does not admit a *categorical semantics* such that Cut is represented by composition of morphisms of some suitable category, as is the case for linear logic (*-autonomous or symmetric monoidal closed categories), intuitionistic logic (cartesian closed categories), and classical logic (control categories). Here we show that NMMS admits a non-trivial interpretation in a different algebraic structure, specifically a Hopf algebra defined over a quotient set of labelled and decorated rooted trees which are just the proof trees of the sequent calculus.

Hopf algebras on rooted trees are well-studied objects that have seen applications in areas ranging from renormalization for perturbative Quantum Field Theory (Connes and Kreimer) to syntactic trees in generative linguistics (Chomsky, Marcolli, Berwick). Hopf algebras for typed and decorated binary trees have also been studied in the combinatorics literature, especially by Foissy. To the author's knowledge no Hopf algebra for proof trees of a sequent calculus has yet been constructed explicitly. The construction developed in this paper does so via the pre-Lie algebra route, and the structural features of NMMS — specifically the absence of weakening — are what make the resulting algebra non-trivial and proof-theoretically meaningful.

The nonmonotonicity property that makes NMMS unamenable to categorical semantics is in fact what makes the Hopf algebra construction non-trivial. The sequent calculus rules naturally admit an interpretation as a *grafting* operation on proof trees – but the absence of Weakening means that the grafting operation is *not* free to arbitrarily extend (or indeed to

contract) contexts by structural rules. This makes the combinatorics of proof tree grafting more constrained and more intricate than the Weakening-admissible setting. In order to define a pre-Lie product directly from proof-tree grafting, we define a proof-theoretically meaningful equivalence relation on iterated graft compositions such that collections of proof trees that are pairwise distinguished only by artificial serialisation of logically concurrent inferences belong to the same equivalence class. Two graftings acting at non-interacting leaf positions may be performed in either order; the sequent calculus records this choice as part of the derivation witness, but neither ordering carries proof-theoretic priority. This is a similar sort of "bureaucratic" redundancy to that which Girard removed in passing from sequent calculus to proof nets, and that Dosen placed at the centre of his analysis of identity of proofs. Without the quotient a naive associator comparison fails — the noise sector contributes asymmetrically across the two bracketings — and the pre-Lie identity cannot be established on raw proof trees, so obstructing the Hopf algebra construction.

The key point is that it is proof-theoretically meaningful quotient itself that determines the relevant rooted trees, the grafting operation, the quotient of bureaucratic symmetries, and ultimately the class-level associator comparison from which the pre-Lie identity emerges.

Two distinct issues then arise. The first is combinatorial: how do iterated graftings decompose? The second is proof-theoretic: which apparent distinctions between ordered pairs of grafts are genuine and which are merely bureaucratic presentation artefacts? The central contribution of the paper is to show that these two questions can be separated cleanly. The geometry of a graft pair yields a decomposition into independent and nested cases. Passing to a quotient of raw graft-pair data isolates the quotient-trivial overlap sector, leaves a rigid residual sector, and makes the associator comparison transparent.

The argument proceeds in three stages. First, one shows that every graft pair is governed by a simple rooted-tree geometry: the two insertions are either nested or independent. Second, one passes from raw graft-pair witnesses to a quotient that identifies the genuinely bureaucratic symmetries of proof composition. This separates a quotient-trivial noise sector from a residual sector. Third, one proves that the mixed residual case is eliminated, so that the residual contribution classes become rigid enough to be matched across the two bracketings of the associator.

Relation to substructural logics. It is worth distinguishing the non-monotonicity studied here from the related but different phenomenon in

linear logic. Linear logic exhibits what one might call *inner* nonmonotonicity: the inference connector $\multimap$ is not monotone, in the sense that provability of $\vdash A \multimap B$ does not imply provability of $\vdash X \otimes A \multimap B$. However, the *provability relation* of linear logic is itself monotone in the meta-sense: adding new axioms or rules cannot destroy existing proofs. The present setting is different: the primitive relation $\mid\!\sim_0$ is allowed to be nonmonotone in the *meta* sense, so that provability itself may be defeated by enlarging the premiss set. This is the stronger, logical notion of nonmonotonicity studied in the literature on defeasible and nonmonotonic reasoning, and it is what makes the absence of weakening proof-theoretically essential rather than merely convenient.

The structural rules programme. The present paper establishes the base case of a broader research programme. The sequent calculus $NMMS(\mid\!\sim_0)$ is currently formulated with multisets (so exchange is built in) and without weakening or contraction. A natural next question is: which structural rules obstruct the existence of a compatible product and coproduct, and hence of a Hopf algebra on proof trees? Adding weakening would make the sequent interfaces non-rigid, potentially destroying the exactness of the leaf-matching constraint on which the coproduct relies. Adding contraction would identify multisets differing in formula multiplicities, changing the quotient structure in ways that may break the noise/residual separation. This programme — using Hopf-algebraic compatibility as a diagnostic for structural rules — is left for subsequent work, but it provides the conceptual frame within which the present results should be read.

Main results. The central results of the paper are the following.

1. **Pre-Lie identity** (Theorem 30). The grafting product $\triangleright$ on $V = \mathbb{Z}\langle PTree\rangle$ satisfies

$$(x \triangleright y) \triangleright z - x \triangleright (y \triangleright z) = (x \triangleright z) \triangleright y - x \triangleright (z \triangleright y)$$

   for all proof trees x, y, z. Hence $(V, \triangleright)$ is a right pre-Lie algebra.

2. **Hopf algebra on proof trees** (Corollary 37). Via the Oudom–Guin construction, the pre-Lie algebra $(V, \triangleright)$ determines a connected graded Hopf algebra on the symmetric algebra $S(V) \cong \mathbb{Z}[Forest(PTree)]$. Its product extends proof-tree grafting to forests; its coproduct is $\Delta(t) = t \otimes 1 + 1 \otimes t$ on primitive generators $t \in V$; and its antipode is given by the standard connected-graded recursion.

3. **Lie algebra of asymmetric proof substitution** (Corollary 32). The commutator $[x, y] = x \triangleright y - y \triangleright x$ defines a Lie bracket on V, measuring the directional asymmetry of proof substitution.

Lean formalization. The main results have been formalized in Lean 4 using Mathlib. The central theorem `classLevel_preLie_identity` is machine-verified with no `sorry` placeholders in the live proof chain.

The paper is organized as follows. Section 2 defines the primitive consequence relation and the calculus $NMMS(\mid\!\sim_0)$. Section 3 develops proof trees as rooted ordered objects. Section 4 introduces matching-leaf grafting. Section 5 analyses the two-step graft geometry underlying the associator. Section 6 proves the pre-Lie identity. Section 7 develops the Oudom–Guin Hopf algebra. Section 8 discusses the admissible-cut coproduct as complementary structure. Section 9 records theorem targets and the structural-rules programme.

2 Primitive consequence and induced derivability

2.1 Base vocabulary and full language

Let L_0 be a set of atomic sentence letters. We set

$$L_B := L_0 \cup \{\bot\},$$

where $\bot$ is treated as a primitive incompatibility marker.

Definition 1. *The full language L is generated inductively from the base fragment L_B by*

$$A ::= p \mid \bot \mid (A \Rightarrow A) \mid (A \wedge A) \mid (A \vee A) \mid \neg A, \qquad p \in L_0.$$

Remark 2. *At the base level, $\bot$ is not governed by any special elimination rule. It functions simply as a primitive marker of incompatibility. In particular, sequents such as $A, B, C \mid\!\sim_0 \bot$, with $A, B, C \in L_0$, may be used to record material incompatibility without imposing any general explosion principle.*

2.2 Primitive consequence

Definition 3. *A* primitive consequence relation *is a binary relation*

$$\mid\!\sim_0 \subseteq \ MSet(L_B) \times MSet(L_B),$$

where $MSet(L_B)$ denotes the finite multisets of formulas of the base fragment L_B.

Remark 4. *No structural conditions are imposed on $\mid\!\sim_0$. In particular, from $\Gamma \mid\!\sim_0 \Theta$ one cannot in general infer $(\Gamma, A) \mid\!\sim_0 \Theta$ or $\Gamma \mid\!\sim_0 (\Theta, A)$. Primitive consequence is given only on the base fragment L_B; the logical vocabulary of L is added above it by the sequent rules.*

2.3 Sequents

A *sequent* is an expression $\Gamma \mid\!\sim \Delta$ where Γ and Δ are finite multisets of L-formulas. We write $Seq(L)$ for the set of all such sequents.

2.4 The calculus $NMMS(\mid\!\sim_0)$

We now define the nonmonotonic multi-succedent sequent calculus generated by $\mid\!\sim_0$.

Definition 5. *A* derivation *in $NMMS(\mid\!\sim_0)$ is an element of the least class closed under the following formation clauses.*

Base axiom. Whenever $\Gamma, \Theta \in MSet(L_B)$ and $\Gamma \mid\!\sim_0 \Theta$, we may form the sequent

$$\frac{\Gamma \mid\!\sim_0 \Theta}{\Gamma \mid\!\sim \Theta} \quad \textit{(Base).}$$

Implication rules.

$$\frac{\Gamma \mid\!\sim A, \Theta \qquad B, \Gamma \mid\!\sim \Theta}{(A \Rightarrow B), \Gamma \mid\!\sim \Theta} \quad (\Rightarrow L)$$

$$\frac{A, \Gamma \mid\!\sim B, \Theta}{\Gamma \mid\!\sim (A \Rightarrow B), \Theta} \quad (\Rightarrow R)$$

Conjunction rules.

$$\frac{A, B, \Gamma \mid\!\sim \Theta}{(A \wedge B), \Gamma \mid\!\sim \Theta} \quad (\wedge L)$$

$$\frac{\Gamma \mid\!\sim A, \Theta \qquad \Gamma \mid\!\sim B, \Theta}{\Gamma \mid\!\sim (A \wedge B), \Theta} \quad (\wedge R)$$

Disjunction rules.

$$\frac{A, \Gamma \mid\!\sim \Theta \qquad B, \Gamma \mid\!\sim \Theta}{(A \vee B), \Gamma \mid\!\sim \Theta} \quad (\vee L)$$

$$\frac{\Gamma \mid\!\sim A, B, \Theta}{\Gamma \mid\!\sim (A \vee B), \Theta} \quad (\vee R)$$

Negation rules.

$$\frac{\Gamma \mid\!\sim A, \Theta}{\neg A, \Gamma \mid\!\sim \Theta} \quad (\neg L) \qquad \frac{A, \Gamma \mid\!\sim \Theta}{\Gamma \mid\!\sim \neg A, \Theta} \quad (\neg R)$$

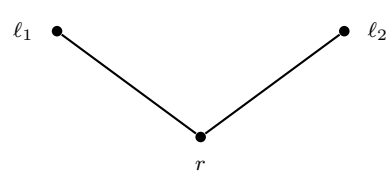
Remark 6. *The logical rules are standard active-formula rules. The important structural choice is not the shape of these rules but the absence of built-in structural principles such as weakening. This is the proof-theoretic setting in which the primitive relation $\mid\!\sim_0$ genuinely matters.*

Remark 7. *The logical rules of NMMS($\mid\!\sim_0$) do not generate inferential content from nothing: they provide vocabulary for expressing and articulating the entailment structure already present in $\mid\!\sim_0$. The passage $\mid\!\sim_0 \rightsquigarrow \vdash_{NMMS(\mid\!\sim_0)}$ is the transition from primitive material consequence to its explicit logical articulation. The absence of weakening is not a restriction imposed on an otherwise unconstrained system; it is the condition that prevents the logical vocabulary from overwriting or silently extending the primitive inferential structure it is meant to express.*

The following figures display a small derivation both proof-theoretically and as a labelled rooted tree, and then show the grafting operation.

$$\dfrac{\dfrac{\Gamma \mid\!\sim_0 (A \Rightarrow B), \Theta}{\Gamma \mid\!\sim (A \Rightarrow B), \Theta}\,\textsc{Base} \qquad \dfrac{\Gamma \mid\!\sim_0 C, \Theta}{\Gamma \mid\!\sim C, \Theta}\,\textsc{Base}}{\Gamma \mid\!\sim ((A \Rightarrow B) \wedge C), \Theta}\,\wedge R$$

The same derivation may also be displayed in graph-theoretic form as an ordinary rooted ordered tree. To match the proof-theoretic orientation above, we place the root at the bottom and draw edges upward toward the premisses.



$$
\begin{aligned}
\rho(r) &= \wedge R, \qquad \sigma(r) = \Gamma \mid\!\sim ((A \Rightarrow B) \wedge C), \Theta, \\
\rho(\ell_1) &= \textsc{Base}, \qquad \sigma(\ell_1) = \Gamma \mid\!\sim (A \Rightarrow B), \Theta, \\
\rho(\ell_2) &= \textsc{Base}, \qquad \sigma(\ell_2) = \Gamma \mid\!\sim C, \Theta.
\end{aligned}
$$

Formally, this is the same proof object viewed as a finite rooted ordered tree (V, E, r) together with a sequent-labelling map $\sigma : V \to \mathit{Seq}(L)$ and a rule-labelling map

$$\rho : V \to \{\mathsf{Base}, \Rightarrow L, \Rightarrow R, \wedge L, \wedge R, \vee L, \vee R, \neg L, \neg R\}.$$

If v is an internal vertex with ordered children $v_1, \ldots, v_n$, then $\sigma(v_1), \ldots, \sigma(v_n)$ are exactly the premisses of an instance of the rule $\rho(v)$ with conclusion $\sigma(v)$. If v is a leaf, then $\rho(v) = \mathsf{Base}$, and $\sigma(v)$ is justified by a primitive consequence instance.

2.5 Induced derivability

Definition 8. *For $\Gamma, \Delta \in \mathit{MSet}(L)$, we write*

$$\Gamma \vdash_{\mathit{NMMS}(\mid\!\sim_0)} \Delta$$

if there exists a derivation of the sequent $\Gamma \mid\!\sim \Delta$ in the calculus $\mathit{NMMS}(\mid\!\sim_0)$.

Remark 9. *The distinction between the primitive $\mid\!\sim_0$ and the induced derivability $\vdash_{\mathit{NMMS}(\mid\!\sim_0)}$ is mathematically significant in the nonmonotonic setting: the latter is not in general a monotone extension of the former. The passage $\mid\!\sim_0 \rightsquigarrow \vdash_{\mathit{NMMS}(\mid\!\sim_0)}$ is the proof-theoretic closure from primitive material consequence to full derivability.*

3 Proof trees

3.1 Tree representation

Definition 10. *A* proof tree *is a tuple*

$$t = (V_t, E_t, r_t, \lambda_t, \sigma_t),$$

where (V_t, E_t, r_t) is a finite rooted ordered tree, $\sigma_t : V_t \to \mathit{Seq}(L)$ is a sequent-labelling, and

$$\lambda_t : V_t \to \{\mathsf{Base}, \Rightarrow L, \Rightarrow R, \wedge L, \wedge R, \vee L, \vee R, \neg L, \neg R\},$$

such that:

(i) *if v is a leaf, then $\lambda_t(v) = \mathsf{Base}$;*

(ii) *if v is an internal vertex, then $\lambda_t(v) \neq \mathsf{Base}$, and the ordered immediate subtrees of v are exactly the premise trees of an instance of the rule $\lambda_t(v)$ with conclusion equal to the sequent label of v.*

The sequent at the root is called the conclusion *of the proof tree.*

Remark 11. *The order on immediate subtrees is part of the formal object, which is why addresses are later indexed by finite sequences of child indices. Conceptually, these are ordinary rooted proof trees.*

Proposition 12. *Every derivation d in NMMS($\mid\!\sim_0$) determines a proof tree Tree(d) with the same conclusion sequent.*

Proof. We argue by induction on the height of the derivation d.

If d has height 0, then d is a base derivation

$$\frac{\Gamma \mid\!\sim_0 \Theta}{\Gamma \mid\!\sim \Theta}.$$

We assign to d the one-vertex rooted tree whose unique vertex is labelled by the sequent $\Gamma \mid\!\sim \Theta$ and tagged by Base. Its root label is therefore the conclusion sequent of d.

Suppose now that d ends with an application of one of the logical rules of *NMMS*($\mid\!\sim_0$), with immediate subderivations $d_1, \ldots, d_n$. By the induction hypothesis, each d_i determines a finite rooted ordered proof tree *Tree*(d_i) whose root carries the conclusion sequent of d_i. Form a new rooted ordered tree by adjoining a fresh root r, declaring the trees *Tree*(d_1), ..., *Tree*(d_n) to be the ordered immediate subtrees of r, labelling r by the conclusion sequent of the last inference of d, and tagging r by the rule symbol used in that inference.

Because the premiss sequents of the last rule instance are exactly the conclusion sequents of the subderivations $d_1, \ldots, d_n$, the resulting rooted ordered tree satisfies the defining conditions of a proof tree. It is finite because it is obtained by adjoining one new root to finitely many finite trees. Its root sequent is, by construction, the conclusion sequent of d.

This completes the induction. □

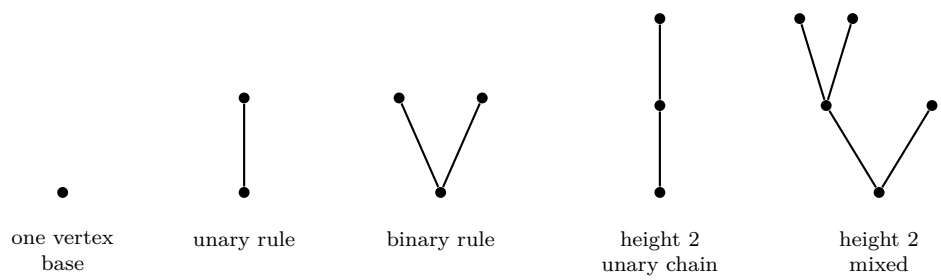
Definition 13 (The set of proof trees)**.** *We write PTree for the set of all finite rooted ordered trees equipped with the rule labels and sequent labels induced by derivations of NMMS($\mid\!\sim_0$). Thus an element $t \in$ PTree may be written as*

$$t = (V_t, E_t, r_t, \lambda_t, \sigma_t),$$

where V_t is the finite vertex set, $E_t \subseteq V_t \times V_t$ is the edge relation directed away from the root r_t, λ_t records the rule label at each vertex, and σ_t records the sequent label. When the labels play no explicit role we suppress them and write simply $t = (V_t, E_t, r_t)$.

Remark 14. *All trees occurring below are elements of PTree. In particular, the raw grafting product and its linearization are defined on PTree itself; the quotient enters later at the level of* two-step witnesses *for the associator, not at the level of the underlying one-step product.*

Small shapes. As in the usual Hopf algebras of rooted trees, it is helpful to keep the first few shapes in view. The following pictures suppress sequent labels and rule tags; they show only the underlying rooted ordered tree. In our setting the allowed labels on the root determine the arity: base axioms give leaves, unary logical rules give one-child nodes, and binary logical rules give two-child nodes.



For a fixed number of vertices there are only finitely many underlying ordered shapes, but each shape supports many proof trees because its vertices may be decorated by different sequents and rule tags. The algebra below is therefore not the ordinary Grossman–Larson algebra on bare trees; it is the same kind of rooted-tree combinatorics carried by proof-theoretic decorations and exact sequent interfaces.

3.2 Rule arity, height, and proof-tree induction

The rules of $NMMS(\mid\!\sim_0)$ divide by number of premisses: Base has arity 0; the unary rules $\Rightarrow R$, $\wedge L$, $\vee R$, $\neg L$, $\neg R$ have arity 1; and the binary rules $\Rightarrow L$, $\wedge R$, $\vee L$ have arity 2. A proof tree is *arity-correct* if every vertex has exactly as many children as its rule arity demands.

Theorem 15 (Derivation trees respect rule arity)**.** *If d is a derivation of $NMMS(\mid\!\sim_0)$, then the associated proof tree $Tree(d)$ is arity-correct.*

Proof. The proof is by induction on the final rule of d. A base derivation is sent to a one-vertex tree, and $\alpha(\mathsf{Base}) = 0$. If d ends with a logical rule R with premisses $d_1, \ldots, d_n$, then the construction of $\mathit{Tree}(d)$ adjoins a new root whose children are exactly $\mathit{Tree}(d_1), \ldots, \mathit{Tree}(d_n)$. By the definition of the inference rules, $n = \alpha(R)$, and by the induction hypothesis the children are themselves arity-correct. □

Definition 16 (Tree height)**.** *The* height $h(t)$ *of a proof tree is defined recursively by*

$$h(t) = 0 \quad \textit{if } t \textit{ is a leaf},$$

and, if t has immediate subtrees $t_1, \ldots, t_n$,

$$h(t) = 1 + \max\{h(t_i) : 1 \leq i \leq n\}.$$

The empty maximum does not occur in the second clause, since leaves are treated by the first clause.

Theorem 17 (Derivation height equals tree height)**.** *For every derivation d,*

$$h(\mathit{Tree}(d)) = \mathrm{height}(d).$$

Proof. Again argue by induction on d. The base case is immediate. In the inductive case, the height of the derivation is one plus the maximum height of its immediate subderivations, and the height of the associated tree is one plus the maximum height of the corresponding immediate subtrees. The induction hypothesis identifies these two maxima. □

Corollary 18 (Induction principles)**.** *Any property $P(t)$ of proof trees which is preserved by adjoining a rule-labelled root above arity-correct child trees may be proved for all derivation trees by induction on derivation height, equivalently by induction on tree height. Concretely, it suffices to prove:*

(i) *$P(t)$ for every base leaf;*

(ii) *for each rule R of arity n, if $P(t_i)$ holds for arity-correct proof trees $t_1, \ldots, t_n$, then $P(R(t_1, \ldots, t_n))$ holds.*

Remark 19. *The induction is not over arbitrary n-ary trees but over rule-labelled trees whose branching is controlled by the logical rule at the root. The cases decompose by inference rule, with the number of children fixed by rule arity. This is the proof-theoretically natural induction measure, and it is what makes the algebraic combinatorics tractable.*

3.3 Addresses

Definition 20. *An* address *in a proof tree is a finite sequence of natural numbers indicating a path from the root by successive child indices. The empty address selects the whole tree.*

Definition 21 (Address set and vertex map)**.** *If $t = (V_t, E_t, r_t)$ is a rooted ordered proof tree, we write*

$$Addr(t) \subseteq \mathbb{N}^{<\omega}$$

for the set of all finite sequences $a = (a_1, \ldots, a_n)$ for which there exists a chain of vertices

$$v_0 = r_t,\ v_1, \ldots, v_n \in V_t$$

such that, for each $i < n$, the vertex v_{i+1} is the a_{i+1}-th element of the ordered child list $\mathrm{ch}_t(v_i)$. The empty sequence $\langle\rangle$ belongs to $Addr(t)$ and names the root. We write

$$v_t : Addr(t) \to V_t$$

for the induced vertex map, sending an address to the corresponding vertex. If $a, c \in \mathbb{N}^{<\omega}$, we write $a{\cdot}c$ for concatenation of addresses. For addresses a, b, we write $a \preceq b$ if a is an initial segment of b, and $a \prec b$ if $a \preceq b$ and $a \neq b$. Thus b extends *a exactly when $a \preceq b$. Two addresses are* incomparable *if neither $a \preceq b$ nor $b \preceq a$.*

Remark 22. *Addresses are not conceptually basic; they are a bookkeeping device for locating graft sites and comparing nested versus parallel graft regimes.*

Definition 23 (Address cones and rooted subtrees)**.** *Let $t \in PTree$ and $a \in Addr(t)$. The* address cone below *a is the principal upset*

$$Addr_t(a) := \{\, b \in Addr(t) \mid a \preceq b \,\}.$$

The rooted subtree of t at a *is denoted $t \downarrow_a$. Its address set is*

$$Addr(t \downarrow_a) := \{\, c \in \mathbb{N}^{<\omega} \mid a \cdot c \in Addr(t) \,\},$$

and its rule and sequent labels are inherited from those of t along the translation $c \mapsto a \cdot c$.

Proposition 24 (Disjointness of incomparable address cones)**.** *Let $t \in PTree$ and $a, b \in Addr(t)$. If a and b are incomparable, then*

$$Addr_t(a) \cap Addr_t(b) = \varnothing.$$

Proof. Suppose $d \in Addr_t(a) \cap Addr_t(b)$. Then $a \preceq d$ and $b \preceq d$. Since a and b are both initial segments of the same finite sequence d, one of them must be an initial segment of the other. Hence $a \preceq b$ or $b \preceq a$, contradicting incomparability. $\square$

Remark 25 (Finiteness of address chains)**.** *For each fixed proof tree t, the set $Addr(t)$ is finite, since the vertex set V_t is finite and the vertex map $v_t : Addr(t) \to V_t$ is bijection onto the vertices of t. Hence $(Addr(t), \preceq)$ is a finite partially ordered set. In particular, every chain*

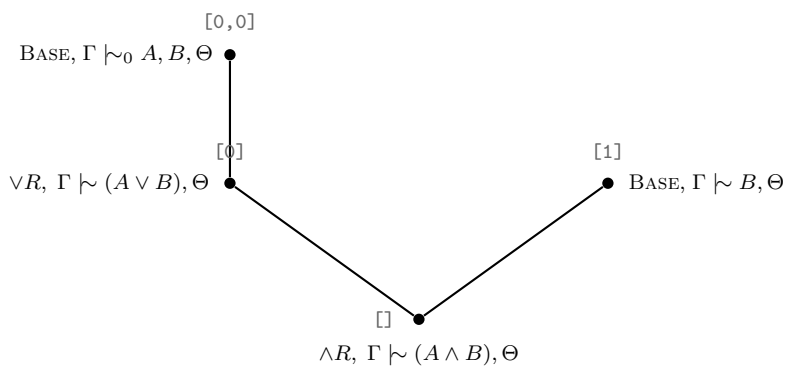
$$a_0 \prec a_1 \prec \cdots \prec a_k$$

of comparable addresses is finite. Thus for any fixed derivation, iterated nesting of graft sites may occur only finitely many times: one can move strictly downward in the tree only along a finite address chain.

Remark 26 (Use of cones in the two-step analysis)**.** *The notations $Addr_t(a)$ and $t\downarrow_a$ are used later in the graft-pair decomposition. If*

$$\mathrm{gr}_a(y, z) = z' \qquad \textit{and} \qquad \mathrm{gr}_b(x, z') = w,$$

then the inner *case is the condition $b \in Addr_{z'}(a)$, i.e. $a \preceq b$, while the* outer *case is the condition that a and b are incomparable in $(Addr(z'), \preceq)$. Proposition 24 is exactly the set-theoretic statement that the rooted regions affected by two outer grafts are disjoint.*



Each node carries a rule tag and a sequent label. Addresses (grey `monospace`) record paths from the root by child index: `[0]` is the first child of the root, `[1]` the second, and `[0,0]` the first child of `[0]`. The nodes at `[0,0]` and `[1]` are leaves (open premiss positions); the node at `[0]` is an internal node with one child.

4 Matching-leaf grafting

The next step is to turn proof substitution into an algebraic operation. The operation is more rigid than ordinary rooted-tree insertion: a tree may be inserted only at a leaf whose sequent label is exactly the conclusion sequent of the tree being inserted. In a system admitting weakening, one could freely enlarge any leaf's context to match an arbitrary insertion target, making all graft sites universally accessible. Here weakening is absent, so context interfaces cannot be stretched: each graft site must be earned by exact sequent agreement. This is the point at which the failure of monotonicity of $\mid\!\sim_0$ becomes directly visible in the algebra, and it is what makes the resulting product structurally non-trivial.

Matching leaves. Let $x, t \in \mathit{PTree}$. We write

$$\mathrm{Leaf}_x(t)$$

for the finite set of leaves $a \in \mathit{Addr}(t)$ whose sequent label is the root sequent of x. For $a \in \mathrm{Leaf}_x(t)$, let

$$\mathrm{gr}_a(x, t)$$

be the proof tree obtained by replacing the leaf a of t by the whole tree x. Since the conclusion of x is exactly the sequent carried by the leaf, this replacement preserves correctness of the surrounding rule instance. Thus $\mathrm{gr}_a(x, t)$ is again a proof tree.

Proposition 27 (Closure under matching-leaf grafting)**.** *If x and t are proof trees of NMMS($\mid\!\sim_0$) and $a \in \mathrm{Leaf}_x(t)$, then $\mathrm{gr}_a(x, t)$ is again the proof tree of a derivation of NMMS($\mid\!\sim_0$).*

Proof. The leaf at a is a base-premiss occurrence whose sequent label agrees with the conclusion of x. Replacing that leaf by the derivation tree x therefore preserves the premiss expected by the parent inference rule. All other vertices and rule instances are unchanged. Reading the resulting tree from leaves to root gives a derivation with the same conclusion as t. □

Linearization. Let $V = \mathbb{Z}\langle \mathit{PTree} \rangle$ be the free abelian group on proof trees. For basis trees define

$$x \triangleright t := \sum_{a \in \mathrm{Leaf}_x(t)} \mathrm{gr}_a(x, t),$$

and extend bilinearly to V. The sum is finite because t has finitely many leaves. This is the proof-theoretic analogue of the Grossman–Larson insertion product: one inserts x into all admissible sites of t, but the admissible sites are determined by exact sequent-interface matching.

Running example. Suppose t has two open leaves carrying the same sequent $\Gamma \mid\!\sim \Delta$, and x is a derivation of $\Gamma \mid\!\sim \Delta$. Then

$$x \triangleright t = \mathrm{gr}_{a_1}(x,t) + \mathrm{gr}_{a_2}(x,t),$$

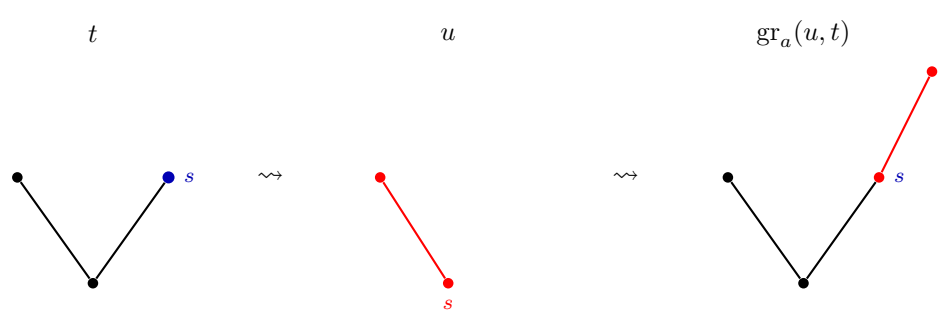
one term for each matching open leaf. If a third leaf differs only by an extra formula in the context, it is not a matching site: without weakening one cannot enlarge or shrink interfaces in order to force a graft.

Here is the same operation with the proof-theoretic labels visible. The left tree t has an open base leaf labelled $\Gamma \mid\!\sim (A \vee B), \Theta$. The middle tree u, drawn in red, derives that exact sequent. Grafting replaces the matching leaf of t by the whole derivation u.

$$\underbrace{\dfrac{\dfrac{\Gamma \mid\!\sim_0 (A \vee B), \Theta}{\Gamma \mid\!\sim (A \vee B), \Theta}\,\textsc{Base} \qquad \dfrac{\Gamma \mid\!\sim_0 C, \Theta}{\Gamma \mid\!\sim C, \Theta}\,\textsc{Base}}{\Gamma \mid\!\sim ((A \vee B) \wedge C), \Theta}\,\wedge R}_{t} \qquad \underbrace{\dfrac{\dfrac{\Gamma \mid\!\sim_0 A, B, \Theta}{\Gamma \mid\!\sim A, B, \Theta}\,\textsc{Base}}{\Gamma \mid\!\sim (A \vee B), \Theta}\,\vee R}_{u}$$

$$\underbrace{\dfrac{\dfrac{\dfrac{\Gamma \mid\!\sim_0 A, B, \Theta}{\Gamma \mid\!\sim A, B, \Theta}\,\textsc{Base}}{\Gamma \mid\!\sim (A \vee B), \Theta}\,\vee R \qquad \dfrac{\Gamma \mid\!\sim_0 C, \Theta}{\Gamma \mid\!\sim C, \Theta}\,\textsc{Base}}{\Gamma \mid\!\sim ((A \vee B) \wedge C), \Theta}\,\wedge R}_{\mathrm{gr}_a(u,t)}$$

The graph-theoretic picture is the same substitution with labels suppressed. The blue vertex marks the matching leaf in t, the red tree is the inserted derivation u, and the right-hand tree shows the inserted branch in red.



5 Associators and the geometry of two grafts

The pre-Lie identity is a statement about associators. In the present setting the relevant combinatorics is not the full syntax of derivations, but the relative position of two graft sites in a rooted tree.

Two-step grafts. Consider a two-step composite

$$\mathrm{gr}_b(x, \mathrm{gr}_a(y, z)).$$

The first graft inserts y into z at a. The second graft inserts x into the resulting tree at b. There are only two possible geometric regimes.

Theorem 28 (Two-step decomposition)**.** *Let $z' = \mathrm{gr}_a(y, z)$, and suppose b is a matching leaf of z' for x. Then exactly one of the following holds.*

(i) Nested case. *The address b lies inside the copy of y inserted at a. Equivalently, $b = a \cdot c$ for a unique nonempty address $c \in Addr(y)$. In this case*

$$\mathrm{gr}_{a\cdot c}(x, \mathrm{gr}_a(y, z)) = \mathrm{gr}_a(\mathrm{gr}_c(x, y), z).$$

(ii) Parallel case. *The addresses a and b are incomparable. In this case the two local substitutions commute:*

$$\mathrm{gr}_b(x, \mathrm{gr}_a(y, z)) = \mathrm{gr}_a(y, \mathrm{gr}_b(x, z)).$$

Proof. The address a becomes the root address of the inserted copy of y in z'. If b extends a, then $b = a \cdot c$ for a unique address c of y. The case $c = \langle\rangle$ would graft x at the root of the inserted copy rather than at a leaf inside it, so for a genuine second graft inside y the extension is nonempty. Locality of replacement gives the displayed nested identity.

If b does not extend a, then b cannot be a strict prefix of a: the address b is a leaf of z', and no valid address can lie strictly below a leaf. Hence a and b are incomparable. The rooted subtrees at incomparable addresses are disjoint, so replacing one leaves the other unchanged. Therefore the two replacements commute. □

Bureaucracy. The same final two-step proof tree may be presented by different ordered histories. In the parallel case, the two histories differ only by the order in which two independent substitutions are performed. In the nested case, one may either first form an inner composite and then insert it, or insert the outer tree first and then work inside it. We write $\sim_{\mathrm{b}}$ for the equivalence relation on two-step graft histories generated by these two local identifications. Each generator preserves the terminal proof tree.

Remark 29. *This is the proof-theoretic analogue of quotienting away bureaucratic permutations of independent inferences. It is close in spirit to the passage from sequent derivations to proof nets, but here the quotient is used only to organize the associator comparison.*

6 The pre-Lie theorem

For $x, y, z \in PTree$, write

$$A(x, y, z) := (x \triangleright y) \triangleright z - x \triangleright (y \triangleright z)$$

for the right associator of the grafting product.

Theorem 30 (Grafting is right pre-Lie)**.** *The bilinear product $\triangleright$ on $V = \mathbb{Z}\langle PTree\rangle$ satisfies*

$$A(x, y, z) = A(x, z, y)$$

for all proof trees x, y, z. Hence V is a right pre-Lie algebra.

Proof. By bilinearity it suffices to compare coefficients of a fixed output tree w. Such a coefficient is counted by two-step graft histories with terminal tree w.

The two-step decomposition separates these histories into nested and parallel regimes. Nested histories are exactly those accounted for by the subtraction term $x \triangleright (y \triangleright z)$: they amount to first inserting x into y, then inserting the result into z. Thus the nested contribution is removed from the associator, leaving only the parallel regime.

The remaining histories are parallel: the two insertions occur at mutually incomparable addresses of the same receiver tree. We pass to a quotient of two-step graft histories, identifying those that differ only by the serialisation of such independent concurrent operations. In each equivalence class under this quotient, one may exchange the roles of y and z without changing the terminal tree: the class governed by inserting y at a and x at a site in the unchanged portion of z corresponds exactly, under commutation at incomparable addresses, to the class in $A(x, z, y)$ governed by the same pair of sites with y and z exchanged.

Concretely: the quotient separates contributions into a *noise sector* — histories that are trivial modulo the parallel-commutativity equivalence and cancel across the two bracketings — and a *residual sector* of genuinely asymmetric contributions. The noise sector contributes equally to $A(x, y, z)$ and $A(x, z, y)$ and so drops out. In the residual sector the exchange of y and z gives an explicit sign-preserving bijection between contribution classes. Therefore the coefficients of every output tree w agree across the two bracketings, and $A(x, y, z) = A(x, z, y)$. □

Remark 31. *The formal development proves this comparison at the level of quotient classes of two-step histories. The quotient is not an extra algebraic carrier; it is the identity criterion used in the associator proof. The one-step product is the raw grafting product on proof trees.*

Corollary 32 (The induced Lie algebra)**.** *The commutator*

$$[x, y] := x \triangleright y - y \triangleright x$$

defines a Lie bracket on V.

Proof. Every right pre-Lie algebra has a Lie algebra commutator. Antisymmetry is immediate, and the Jacobi identity is the standard cancellation obtained by expanding the pre-Lie identity in the three cyclic orders. □

Remark 33. *The Lie bracket $[x, y] = x \triangleright y - y \triangleright x$ has a direct proof-theoretic reading: it measures the* asymmetry of proof substitution*. The term $x \triangleright y$ inserts x into the open premiss positions of y; the term $y \triangleright x$ does the reverse. Their difference records which insertions are available in one direction but not the other — a quantity that is non-trivial precisely because the leaf-matching constraint is sensitive to the direction of substitution. That this difference satisfies the Jacobi identity is a structural theorem about the proof-tree algebra, not merely an algebraic formality.*

7 Relation with rooted-tree Hopf algebras

The construction above is the proof-theoretic analogue of a familiar rooted-tree story. For ordinary rooted trees, insertion at vertices defines a pre-Lie algebra; its commutator is the Grossman–Larson Lie algebra; and the universal enveloping algebra is a Hopf algebra. The proof-theoretic novelty is that the insertion sites are not arbitrary vertices: they are matching premiss-leaves determined by the primitive consequence relation.

Forests and the symmetric algebra. Let $\mathit{Forest}(\mathit{PTree})$ be the free commutative monoid on proof trees, with unit the empty forest 1. The free commutative algebra

$$\mathbb{Z}[\mathit{Forest}(\mathit{PTree})] \cong S_{\mathbb{Z}}(V)$$

is the symmetric algebra on V. A forest is not itself a single proof; it is the algebraic carrier on which the Hopf algebra lives.

Theorem 34 (Oudom–Guin)**.** *Let A be a right pre-Lie algebra. The symmetric algebra $S(A)$ carries an associative product $\star$, determined by the pre-Lie action, for which*

$$\Delta(a) = a \otimes 1 + 1 \otimes a \qquad (a \in A)$$

extends to a coproduct making $S(A)$ a Hopf algebra. This Hopf algebra is canonically isomorphic to the universal enveloping algebra of the Lie algebra associated to A.

The algebraic structure explicitly. Applied to $A = V$, the Oudom–Guin construction gives the following concrete objects. The underlying module is the symmetric algebra

$$S(V) = \mathbb{Z}[\mathit{Forest}(\mathit{PTree})].$$

Its unit is the empty forest 1. The product $\star$ is the unique Oudom–Guin product extending the proof-tree pre-Lie operation: on generators it records insertion/grafting, and on forests it is extended by the recursive Oudom–Guin rule. The coproduct

is the primitive-generator coproduct, extended as an algebra morphism from the commutative forest algebra:

$$\Delta(1) = 1 \otimes 1, \qquad \Delta(x) = x \otimes 1 + 1 \otimes x \quad (x \in V),$$

and therefore, for a forest $x_1 \cdots x_n$,

$$\Delta(x_1 \cdots x_n) = \sum_{I \sqcup J = \{1,\ldots,n\}} \Big(\prod_{i \in I} x_i\Big) \otimes \Big(\prod_{j \in J} x_j\Big).$$

The counit is

$$\varepsilon(1) = 1, \qquad \varepsilon(F) = 0 \quad \text{for every nonempty forest } F.$$

Proposition 35 (The Oudom–Guin bialgebra)**.** *With the product $\star$, unit 1, coproduct Δ, and counit ε above, $S(V)$ is a connected graded bialgebra.*

Proof. Associativity of $\star$ and compatibility of $\star$ with the primitive coproduct are precisely the Oudom–Guin theorem; the pre-Lie identity is the hypothesis needed for that construction. Coassociativity of Δ on generators follows from

$$(\Delta \otimes \mathrm{id})\Delta(x) = x \otimes 1 \otimes 1 + 1 \otimes x \otimes 1 + 1 \otimes 1 \otimes x = (\mathrm{id} \otimes \Delta)\Delta(x),$$

and hence on all forests because Δ is multiplicative. The counit identities are immediate on 1 and on primitive generators, and again extend multiplicatively. The grading is by forest length or, equivalently, by total proof-tree weight; the degree-zero part is $\mathbb{Z} \cdot 1$, so the bialgebra is connected. □

Theorem 36 (Antipode)**.** *The connected graded bialgebra $S(V)$ has a unique antipode S. It is defined recursively by*

$$S(1) = 1, \qquad S(F) = -F - \sum S(F') \star F'',$$

where the sum ranges over the reduced coproduct

$$\widetilde{\Delta}(F) = \Delta(F) - F \otimes 1 - 1 \otimes F = \sum F' \otimes F''.$$

It satisfies

$$m_\star(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m_\star(\mathrm{id} \otimes S)\Delta.$$

Proof. This is the standard antipode construction for connected graded bialgebras. The reduced coproduct of a homogeneous element of positive degree has tensor factors of strictly smaller degree, so the displayed recursion is well-founded on degree. An induction on degree then shows that convolution with the identity gives the unit-counit map on both sides. Uniqueness follows from the same induction: any convolution inverse of the identity must satisfy the recursive formula. □

Corollary 37 (The Oudom–Guin Hopf algebra of proof trees)**.** *The pre-Lie algebra* $(V, \triangleright)$ *of proof trees determines a Hopf algebra on* $S(V)$*. Its primitive generators are the proof trees, and its Lie algebra of primitives is generated by the proof-theoretic commutator* $[x, y] = x \triangleright y - y \triangleright x$*.*

Remark 38 (Proof-theoretic content of the Hopf structure)**.** *Each component of the Hopf algebra has a direct proof-theoretic reading. The* product $\star$ *on* $S(V)$ *is proof composition extended to forests: combining collections of derivations. The* coproduct $\Delta(t) = t \otimes 1 + 1 \otimes t$ *expresses the fact that, in the absence of weakening and contraction, each proof tree is an* atomic *or* primitive *proof-theoretic object: it cannot be split into genuinely independent sub-components by any structural decomposition. This primitivity is a consequence of the non-monotone constraint — it is the absence of structural rules that prevents non-trivial decomposition. The* antipode *provides a formal inverse for proof composition, whose existence follows from the connected graded structure (graded by derivation height). The* Lie bracket $[x, y]$ *records the directional asymmetry of proof substitution described above.*

This construction should be distinguished from the Connes–Kreimer cut coproduct, in which the coproduct is defined by admissible cuts of trees and encodes genuine proof decomposition. The two coproducts are related but distinct. The Oudom–Guin Hopf algebra established here is the algebraic structure that the pre-Lie proof-tree product determines; the cut coproduct and its relation to the primitive-generator coproduct is a separate problem, discussed briefly in the following section.

Remark 39 (What is proved here, and what remains separate)**.** *The pre-Lie theorem supplies a complete input to the standard Oudom–Guin machine, so the Hopf algebra on* $S(V)$ *follows by an established theorem of pre-Lie theory. This does not by itself prove that the admissible-cut coproduct below descends through the same proof-theoretic quotient, nor that the cut coproduct is the coproduct of the Oudom–Guin Hopf algebra. Those are additional comparison and descent problems.*

8 Cut coproducts and the coalgebra problem

There is nevertheless a natural cut coproduct on proof trees. It is useful both for comparison with Connes–Kreimer theory and for isolating the remaining coalgebraic work.

Admissible cuts. An admissible cut of a proof tree t is a finite antichain

$$C \subseteq Addr(t)$$

for the prefix order. The antichain condition says that a path from the root to a leaf meets at most one cut address. Such a cut determines:

$$P_C(t) = \prod_{a \in C} t \downarrow_a, \qquad R_C(t) = \text{the tree obtained by replacing each } t \downarrow_a \text{ by a leaf.}$$

Here $P_C(t)$ is a forest of pruned subtrees, while $R_C(t)$ is the remainder or trunk.

The cut sum. The associated cut coproduct datum is

$$\Delta_{\mathrm{cut}}(t) = \sum_{C \in \mathrm{Adm}(t)} P_C(t) \otimes R_C(t),$$

with the empty cut contributing $1 \otimes t$. Depending on convention, the root cut contributes $t \otimes 1$, or this term is adjoined separately.

Proposition 40 (Node decomposition of cuts)**.** *Let $t = R(t_1, \ldots, t_n)$ be a proof tree whose root is a rule of arity n. Every admissible non-root cut of t is uniquely a tuple of admissible cuts C_i of the immediate subtrees t_i. Under this correspondence,*

$$P_C(t) = P_{C_1}(t_1) \cdots P_{C_n}(t_n), \qquad R_C(t) = R(R_{C_1}(t_1), \ldots, R_{C_n}(t_n)).$$

Proof. Every non-root address of t has a unique first child index i, followed by an address of t_i. Restricting an antichain C to addresses with first index i gives an antichain C_i in t_i. Conversely, the union of the prefixed child cuts is an antichain in t: addresses with different first indices lie in disjoint branches, and addresses with the same first index are controlled by the antichain condition in the corresponding child. The formulas for P_C and R_C follow directly from this decomposition. □

Theorem 41 (Coassociativity criterion)**.** *The coassociativity of Δ_{cut} is equivalent to the usual nested-cut decomposition: a two-stage cut of a proof tree may be read either as cutting first and then cutting the pruned forest, or as cutting the remainder and then collecting the newly pruned components.*

Proof sketch. Both sides enumerate the same finite data: compatible families of cut addresses together with a choice of stage. The antichain condition ensures that distinct cut components are either disjoint or nested in the permitted way. The node decomposition reduces the proof to the immediate subtrees, so one proves the criterion by induction on tree height, with cases indexed by the rule arity at the root. □

Proposition 42 (Raw cut coalgebra, conditional form)**.** *Suppose the admissible-cut operation includes the empty cut and the root cut, and suppose the nested-cut decomposition above holds for all proof trees. Extend Δ_{cut} multiplicatively to forests and define*

$$\varepsilon_{\mathrm{cut}}(1) = 1, \qquad \varepsilon_{\mathrm{cut}}(F) = 0 \quad \textit{for every nonempty forest } F.$$

Then $(\mathbb{Z}[Forest(PTree)], \Delta_{\mathrm{cut}}, \varepsilon_{\mathrm{cut}})$ is a coalgebra.

Proof. The counit equations hold on a tree t because the empty cut contributes $1 \otimes t$ and the root cut contributes $t \otimes 1$, while every other cut contributes a tensor whose two factors are both nonempty and is therefore killed by applying $\varepsilon_{\mathrm{cut}}$ to either side. Since both Δ_{cut} and $\varepsilon_{\mathrm{cut}}$ are extended multiplicatively, this proves the counit laws on all forests.

For coassociativity, both $(\Delta_{\mathrm{cut}} \otimes \mathrm{id})\Delta_{\mathrm{cut}}(t)$ and $(\mathrm{id} \otimes \Delta_{\mathrm{cut}})\Delta_{\mathrm{cut}}(t)$ enumerate two-stage admissible cuts of t. The nested-cut decomposition gives the bijection between the two enumerations, preserving the pruned forest and the final remainder. Hence the two sums agree on each tree, and multiplicativity extends the equality to forests. □

Descent through quotients. If $R \subseteq V$ is a relation submodule, a coproduct on V descends to V/R precisely when R is a coideal:

$$\Delta(R) \subseteq R \otimes V + V \otimes R, \qquad \varepsilon(R) = 0.$$

Thus the direct cut route to a quotient coalgebra requires proving that the proof-theoretic equivalence relations used in the pre-Lie argument are stable under cut decomposition. This is the exact coalgebraic form of the remaining problem.

Remark 43 (Foissy-style perspective)**.** *Foissy's study of rooted-tree comodules suggests a useful interpretation of the cut construction. A finite collection of derivations closed under taking cut remainders and pruned subtrees should behave like a finite-dimensional comodule: iterated cuts filter the space by depth of decomposition, while primitive elements correspond to indecomposable proof-tree contributions. This is not needed for the pre-Lie theorem, but it is a natural source of invariants for the eventual Hopf algebra.*

9 Proof-theoretic meaning: theorem targets and roadmap

The previous sections isolate the part of the construction that is currently mathematically solid: proof trees carry a constrained grafting product, the two-step graft geometry gives a right pre-Lie identity, and the standard Oudom–Guin theorem then supplies a Hopf algebra on the symmetric algebra of proof trees. The purpose of this section is different. It records the larger theorem targets suggested by the construction. These statements are not all proved in the present paper; they are included as a guide to the proof-theoretic interpretation of the Hopf operations and to the next formal steps.

9.1 The refinement functor and weakening

Let SR be a chosen set of structural rules. Write

$$\mathcal{P}_{\mathrm{SR}}$$

for the category, or partial category, whose objects are proof trees in the calculus with structural rules SR, and whose morphisms are proof-tree refinements or substitutions. Write $\mathcal{S}$ for a category of sequents, with morphisms generated by

admissible sequent transformations such as context extension. There is a forgetful map

$$U_{\mathrm{SR}} : \mathcal{P}_{\mathrm{SR}} \longrightarrow \mathcal{S},$$

sending a proof tree to its conclusion sequent. In the language of Mellies–Zeilberger refinement systems, a proof tree is a refinement of its conclusion sequent, and U_{SR} forgets the refinement.

Theorem Target 44 (Weakening as cartesian lifting)**.** *After choosing the precise categories $\mathcal{P}_{\mathrm{SR}}$ and $\mathcal{S}$, left and right weakening are admissible for* SR *if and only if U_{SR} has cartesian lifts over context-extension morphisms in $\mathcal{S}$.*

Proof sketch. A cartesian lift over a context extension

$$\Gamma \mid\!\sim \Delta \longrightarrow \Gamma, A \mid\!\sim \Delta$$

would assign, functorially, to any proof tree of $\Gamma \mid\!\sim \Delta$ a proof tree of $\Gamma, A \mid\!\sim \Delta$ lying over the extended sequent. This is exactly left weakening. The right-sided case is analogous. Conversely, if weakening is admissible, then every such context-extension morphism has a canonical lift obtained by applying weakening to the whole derivation. The universal property of cartesian lift expresses the fact that any other proof over the extended context factors through this weakened proof modulo the chosen identity criterion on proof trees. □

Conjecture 45 (Meta-nonmonotonicity as failure of fibration)**.** *For nonmonotonic base relations $\mid\!\sim_0$, the functor U_{SR} is generally not a fibration over context extensions. The obstruction is precisely the failure of weakening: extending a context can destroy a primitive base inference and hence destroy the proof tree that would be needed for a cartesian lift.*

Remark 46. *This is the categorical form of the point used throughout the grafting construction. Matching-leaf grafting is rigid because proof trees cannot be transported freely along context extensions. If such transport were always available, the exact sequent interface at a grafting leaf would become much less informative.*

9.2 Grafting support and structural rigidity

For proof trees x, t, recall that $\mathrm{Leaf}_x(t)$ is the set of leaves of t whose sequent label exactly matches the conclusion of x. The support of the product $x \triangleright t$ is therefore controlled by this matching set.

Theorem Target 47 (Weakening versus support monotonicity)**.** *Let $\leq_{\mathrm{ctx}}$ denote context extension on sequents. Weakening is admissible if and only if grafting support is monotone under $\leq_{\mathrm{ctx}}$: whenever the conclusion of x extends to the label of a leaf of t, there is an admissible weakened proof tree x' whose conclusion exactly matches that leaf, and hence contributes to $x' \triangleright t$.*

Proof sketch. If weakening is admissible, any derivation x may be transported to match any context extension of its conclusion sequent. Thus a near-match becomes an exact match after applying weakening. Conversely, suppose support monotonicity holds uniformly. Apply it to the one-leaf proof context whose unique leaf carries the extended sequent. The existence of a matching graft then supplies precisely a weakened derivation. The argument must be made separately for left and right context extension. □

Remark 48. *This is one of the cleanest bridges between proof theory and algebra. The algebraic support of the pre-Lie product remembers exactly where weakening is unavailable.*

9.3 Hopf operations as proof-theoretic operations

There are two Hopf-algebraic routes in play. The Oudom–Guin route is already available from the pre-Lie theorem: proof trees are primitive generators and the coproduct is the primitive-generator coproduct on forests. The cut route is more proof-decompositional: the coproduct sums over admissible cuts of a proof tree. The following targets explain what each operation should mean proof-theoretically.

Theorem Target 49 (Coproduct as proof decomposition)**.** *For the admissible-cut coproduct, $\Delta_{\mathrm{cut}}(t)$ is the universal linear record of all one-stage decompositions of the proof tree t into detached subproofs and a remaining proof context:*

$$\Delta_{\mathrm{cut}}(t) = \sum_{C} P_C(t) \otimes R_C(t).$$

Coassociativity is exactly the statement that staged proof decomposition is independent of whether one first decomposes the detached subproofs or first decomposes the remaining trunk.

Proof sketch. The two sides of the coassociativity equation enumerate the same combinatorial object: a proof tree together with a compatible two-stage family of cut addresses. One side records it as "cut, then cut the forest of detached pieces"; the other records it as "cut, then cut the remaining trunk". The antichain condition on admissible cuts ensures that these staged cuts are compatible. A bijection between the two enumerations gives coassociativity. □

Theorem Target 50 (Antipode as proof-theoretic Mobius inversion)**.** *For a connected cut Hopf algebra of proof trees, the antipode is the inclusion-exclusion operator over proper proof decompositions:*

$$S(t) = -t - \sum_{C \neq \varnothing, \mathrm{root}} S(P_C(t))\, R_C(t).$$

It cancels every nontrivial decomposition of t:

$$m(S \otimes \mathrm{id})\Delta_{\mathrm{cut}}(t) = 0 = m(\mathrm{id} \otimes S)\Delta_{\mathrm{cut}}(t)$$

for every nonempty proof tree t.

Proof sketch. This is the standard connected graded Hopf-algebra recursion, but its interpretation is proof-theoretic. The reduced coproduct lists all proper ways of viewing t as a trunk with detached subproofs. The antipode alternates over these decompositions so that, after multiplying back, every properly decomposable contribution cancels. Thus $S(t)$ is not literally cut elimination; it is algebraic subtraction of all decomposable proof contexts. □

Theorem Target 51 (Primitive elements as indecomposable proof content)**.** *In the cut Hopf algebra, primitive elements are exactly the linear combinations of proof trees whose proper decomposition terms cancel:*

$$\widetilde{\Delta}(p) = 0.$$

Thus primitive elements are the renormalized indecomposable proof contents, not necessarily individual proof trees.

Proof sketch. By definition, p is primitive precisely when $\Delta(p) = p \otimes 1 + 1 \otimes p$. Equivalently, all proper cut terms vanish in the reduced coproduct. Since proper cuts represent nontrivial proof decompositions, the primitive condition says exactly that no proper decomposition remains after linear cancellation. □

Remark 52. *The Oudom–Guin antipode has a different but related reading. In the universal enveloping algebra of the Lie algebra of proof trees, the antipode reverses composition histories:*

$$S(x_1 \star \cdots \star x_n) = (-1)^n x_n \star \cdots \star x_1$$

up to the usual convention for the enveloping product. This suggests an "inverse proof-history" interpretation for the Oudom–Guin Hopf algebra, whereas the cut antipode suggests a Mobius-inversion interpretation over proof decompositions.

9.4 Structural rules as algebraic diagnostics

The construction suggests that structural rules should control algebraic features of the resulting Hopf object. The following table is a programme, not yet a theorem in full generality.

Structural rule	Categorical reading	Expected algebraic effect
Exchange	symmetry of tensor/order	cocommutativity or symmetric quotient
Weakening	cartesian lifting of U_{SR}	counit/descent from sequent level
Contraction	diagonal $A \to A \otimes A$	divided-power or diagonal coproduct terms
Cut	composition of proofs	nontrivial grafting/pre-Lie product

Conjecture 53 (Functoriality in structural rules)**.** *If* $\mathrm{SR} \subseteq \mathrm{SR}'$*, then there is a natural morphism of pre-Lie or Hopf-type structures*

$$H_{\mathrm{SR}} \longrightarrow H_{\mathrm{SR}'}$$

induced by viewing every proof tree valid with rules SR *as a proof tree valid with rules* SR$'$*. The family* SR $\mapsto H_{\mathrm{SR}}$ *should form a diagram of Hopf algebras or Hopf-like objects.*

Proof sketch. The inclusion of structural rules gives an inclusion of derivation systems. The nontrivial point is compatibility with the quotient identifying bureaucratic proof histories. If the generating equivalences for SR are preserved by the larger calculus, then the inclusion of proof trees induces a morphism of the corresponding pre-Lie quotients. The Oudom–Guin construction is functorial in pre-Lie maps, giving the Hopf algebra morphism. □

9.5 Deduction-detachment and the nonmonotonic conditional issue

The conditional $\Rightarrow$ in the present calculus is detachable in the ordinary sequent-calculus sense. The nonmonotonicity lives in the base relation $\mid\!\sim_0$, not in a special object-language connective for nonmonotonic implication.

Theorem Target 54 (Deduction-detachment)**.** *For the induced derivability relation,*

$$\Gamma \mid\!\sim A \Rightarrow B, \Delta \quad \Longleftrightarrow \quad A, \Gamma \mid\!\sim B, \Delta.$$

Proof sketch. The right-to-left direction is the rule $(\Rightarrow R)$. The left-to-right direction is height-preserving invertibility of $(\Rightarrow R)$, proved by induction on the height of a derivation. The base relation $\mid\!\sim_0$ does not interfere with this rule-level argument, because base sequents are leaves. □

Remark 55. *This does not collapse the system into classical monotonic consequence. There is no object-language connective $\models\!\Rightarrow$ representing the primitive nonmonotonic relation itself. Attempting to add such a connective with both a full deduction theorem and modus ponens is exactly the setting in which Makinson–Gabbay style collapse phenomena arise. The present system chooses the other route: keep $\Rightarrow$ detachable, and locate nonmonotonicity in the base relation $\mid\!\sim_0$.*

Theorem Target 56 (Failure of cautious monotonicity and cumulative transitivity)**.** *In general, the induced consequence relation need not satisfy cautious monotonicity or cumulative transitivity. These failures are inherited from the failure of monotonicity at the base relation.*

Proof sketch. Choose a base relation with a Tweety-style pattern:

$$B \mid\!\sim_0 F, \qquad B, P \not\mid\!\sim_0 F.$$

Since base sequents embed as derivation leaves, the failure persists at the derived level unless repaired by an explicit logical derivation. Analogous base patterns show the failure of cautious monotonicity and cumulative transitivity. The exact conservativity theorem required here is that the logical extension introduces no new purely base sequents beyond those forced by the rules. □

Remark 57. *This is algebraically relevant because failure of these monotonicity principles is what keeps sequent interfaces rigid. The pre-Lie product is proof composition at exact matching interfaces; if explicating a consequence as a premiss were globally harmless, the matching condition would lose much of its force.*

9.6 Toward $*$-autonomous categories

Blute's theorem on Hopf algebras and ribbon-twisted modules suggests a possible route from proof-tree Hopf algebras to categorical models of linear logic. Very roughly, one would like a chain

$$\mathrm{SR} \longmapsto L_{\mathrm{SR}} \xrightarrow{\text{Oudom–Guin}} H_{\mathrm{SR}} \longmapsto \mathrm{RTMod}(H_{\mathrm{SR}}),$$

where L_{SR} is the proof-tree pre-Lie algebra and $\mathrm{RTMod}(H_{\mathrm{SR}})$ is a ribbon-twisted module category.

Problem 58 (Blute route)**.** *Determine structural-rule hypotheses under which the Oudom–Guin Hopf algebra H_{SR} has the additional ribbon or involutive-antipode structure required for* $\mathrm{RTMod}(H_{\mathrm{SR}})$ *to be cyclic $*$-autonomous.*

Remark 59. *This should not be assumed automatically. The Oudom–Guin construction gives a Hopf algebra, but Blute-style $*$-autonomous representation categories require additional structure. The plausible expectation is that exchange, weakening, and contraction each control part of this extra structure.*

Problem 60 (Hopf monad route)**.** *Find a Hopf monad T on a base $*$-autonomous category such that the algebras or modules of T encode proof-tree composition for NMMS($\mid\!\sim_0$). In such a formulation, the monad unit would correspond to one-node proof trees and monad multiplication to grafting.*

Remark 61. *This would bypass the need to first construct a separate representation category of a Hopf algebra. The Hopf-monad axioms would become the categorical form of the same coherence proved combinatorially by the pre-Lie identity.*

9.7 Livernet rigidity as a diagnostic

Livernet's rigidity theorem for pre-Lie algebras says, roughly, that a connected NAP coalgebra satisfying the appropriate distributive law is free as a pre-Lie algebra. Our proof-tree pre-Lie algebra is not expected to be free: the sequent-matching constraint should impose genuine algebraic restrictions.

Problem 62 (Non-freeness of the proof-tree pre-Lie algebra)**.** *Decide whether the proof-tree pre-Lie algebra admits a NAP coproduct Δ_{NAP} satisfying a Livernet-style distributive law*

$$\Delta_{\mathrm{NAP}}(x \triangleright y) = x \otimes y + \Delta_{\mathrm{NAP}}(x) \triangleright y.$$

If no such coproduct exists, then the failure is evidence that the sequent-matching constraint is algebraically visible.

Possible strategy. Compare the free rooted-tree pre-Lie algebra, where grafting is allowed at all vertices, with the proof-tree algebra, where grafting is allowed only at matching leaves. A Livernet coproduct would force freeness. Therefore one should look for a small labelled proof-tree example in which the number of eligible graft sites depends on $\mid\!\sim_0$ in a way incompatible with the universal free pre-Lie count. Such an example would witness non-freeness. □

Remark 63. *This is a particularly promising test because it relates the algebra directly back to the proof theory. If the algebra were free, the base consequence relation would have disappeared from the pre-Lie structure. Non-freeness is therefore not a defect; it is a sign that the proof-theoretic constraints are still present.*

10 Conclusion

We have constructed a proof-theoretic route from a primitive consequence relation to a class-level pre-Lie identity on proof-tree composition. Formally, the route is

$$\begin{aligned} \mid\!\sim_0 &\longrightarrow \mathit{NMMS}(\mid\!\sim_0) \\ &\longrightarrow \text{proof trees} \longrightarrow \text{grafting geometry} \\ &\longrightarrow \text{graft-pair quotient} \longrightarrow \text{class-level pre-Lie identity.} \end{aligned}$$

The distinctive point is not merely that rooted trees appear, but that they appear as proof objects carrying exact sequent interfaces. Grafting is therefore proof composition, not arbitrary combinatorial insertion. The graft-pair quotient then isolates what should count as invariant in proof composition: independent commutations and quotient-trivial overlap disappear as bureaucracy, while the residual sector becomes pure and rigid enough to satisfy the class-level pre-Lie identity.

From the standpoint of logic, this means that a nonmonotonic proof theory can carry nontrivial algebraic structure without first collapsing to a monotone or structurally saturated setting. From the standpoint of algebra, it shows that pre-Lie behavior can emerge from proof composition once identity of proof is handled carefully at the level of graft-pair interactions. The Hopf-theoretic extensions suggested by the present development are important, but they are best seen as a next step built on top of the logic result already proved here: after quotienting bureaucratic symmetries, proof composition on these nonmonotonic proof trees satisfies a class-level pre-Lie identity.

The quotient is not a stylistic choice. The direct approach to the pre-Lie identity — proving that the raw witness counts balance on both sides of the associator — was attempted and did not close. The equivalence relation on graft-pair codes, and in particular the `outer_comm_inner` identification of comparable-address graftings, is the structure that makes the proof work. This identification has a proof-theoretic reading: it says that two derivation histories for the same output proof tree, which look geometrically different at the address level, should count as the same composition when the relevant graft sites stand in a dependency relation. The noise/residual separation is then a precise account of which histories are genuinely

the same (noise, canceling from the associator) and which are genuinely distinct (residual, matched uniquely across the two bracketings).

The central result is fully machine-verified in Lean 4. The class-level pre-Lie identity depends only on the standard axioms of Lean's foundational system. The broader programme — determining which structural rules obstruct Hopf-algebraic compatibility, and whether the resulting structure encodes specific properties of the primitive consequence relation $\mid\!\sim_0$ — remains open and will be pursued in subsequent work.